Question	Answer	Marks
8(a)	packet / quantum of <u>energy</u>	**M1**
	of electromagnetic radiation	**A1**
8(b)(i)	$E = c^2\Delta m$	**C1**
	$\Delta m = (4.274 \times 10^6 \times 1.60 \times 10^{-19}) / (1.66 \times 10^{-27} \times (3.00 \times 10^8)^2)$ (= 0.00458 u)	**C1**
	$m = 233.915174 + 4.000407 + 0.00458$ = 237.92016 u	**A1**
8(b)(ii)	$E = hc / \lambda$ **or** $E = hf$ **and** $c = f\lambda$	**C1**
	$(4.274 - 4.200) \times 1.60 \times 10^{-13} = (6.63 \times 10^{-34} \times 3.00 \times 10^8) / \lambda$	**C1**
	$\lambda = 1.7 \times 10^{-11}$ m	**A1**
8(b)(iii)	(true) energy of gamma photon is smaller so (true) wavelength is larger	**B1**
8(c)	(anti)neutrinos are emitted during beta decay	**B1**
	particles emitted during beta decay carry varying amounts of energy, so energy of gamma photon is also variable (between decays)	**B1**

Question	Answer	Marks